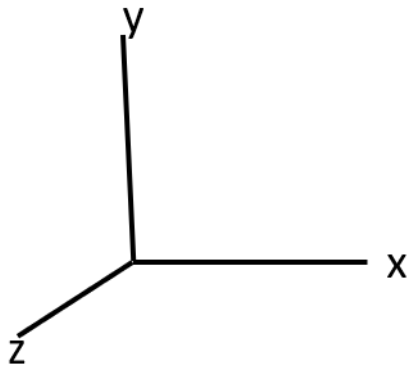
Q1. Camera 5x5 = 25 points

(a) In the Assignment 1, Q1, find the coordinates of the centre of the swinging load, $o_1(x, y, z)$ for all frames in training set rgb_train_val_swingingload.avi. Use the M_{int} obtained in the Lab. Plot the 3D coordinates of swinging load for all frames.



(b) Say, the default intrinsic matrix of L515 is

$$M_d = \begin{bmatrix} 1342 & 0 & 325 \\ 0 & 1342 & 235 \\ 0 & 0 & 1 \end{bmatrix}$$

Use this matrix to find the 3D coordinates of the center of swinging load for all frames (o_2) and plot them.

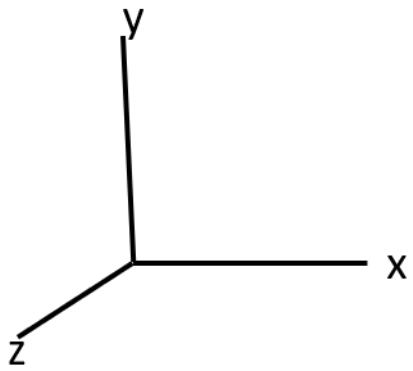
(c) Which intrinsic matrix - $M_{int}\ or\ M_d$ is accurate? Give reasons for your answers.

(d) From the data set obtained from (a) and (b) i.e. $o_1\ and\ o_2$. Implement ML model that learns the function parameters such that $f_\theta(o_1) = o_2^*$. Explain the results by finding/demonstrating the relevant KPIs.

(e) Find the Euclidean error between (i) $o_1 and\ o_2$ and (ii) $o_1 and\ o_2^*$. Make the interpretations of the results that you obtain.

Q2. Lidar 10+15 = 25 points

(a) Use Ouster lidar in Lab to record the trajectory of swinging load. Find the center of swinging load for all frames, and plot them as shown below



(b) Leverage KISS ICP to find the pose of center of load in consecutive lidar frame. **You can use the off the shelf KISS-ICP algorithm, however, It is expected that student understand KISS ICP, its working principle, uses and drawbacks, and present during the presentations. Further present the pose of object obtained from KISS-ICP from the data in dataset above.**

Q3. Camera-Lidar fusion 10 + 15 = 25 points

KITTI data set is comprehensive data set used for benchmarking different algortithms. The home page of KITTI data set is here.

As this assignment has large data set e.g. KITTI data set , so I suggest that students use use UiA's high performance computing and server - coder.internal.uia.no/. The details of how to setup the server is explained in Modules\MachineLearning\Lab-ML, section – Object Detection – YOLO

Lab - ML

ML-Module has following lab:

- Regression Regression_class.rar
  - Follow instructions mentioned in regression_class.ipynb
  - Students are expected to complete the assigned tasks

- Classification
  - Binary classification use case- Motor bearing fault classification usign ML Classifiction-fault.rar
    - This folder contains ./images_train_val_Cropped --> images for classification task
    - fault-classification-class.ipynb --> contains the template, examples to perform the classification task. Further some tasks are a

- Transfer learning, Fine tuning, Feature fusion classification-transferlearning-tuning.rar
  - This folder contains ./classification-transferlearning-tuning-class.ipynb. It contains the example for tasks - transfer learning, fine tu
  - Students are expected to complete the task assigned at the last cell.

- Object Detection - YOLO Custom-objectDetection - tracking -YOLO-class.rar
  - Create a virtual environment in coder using requirements-ultralytics-yolo.txt (list of dependencies to make virtual environment). L UiA (https://www.uia.no/uia-hjelp/it/vpn.html). Follows the instruction in installing coder-server.docx
  - Custom object detection using YOLO
    - Follow the instructions on tutorial - https://youtu.be/wuFCDEt2zIQ?list=PL_3KtWn2bBtFcDqXUTd44C_NWfjIaI1sI

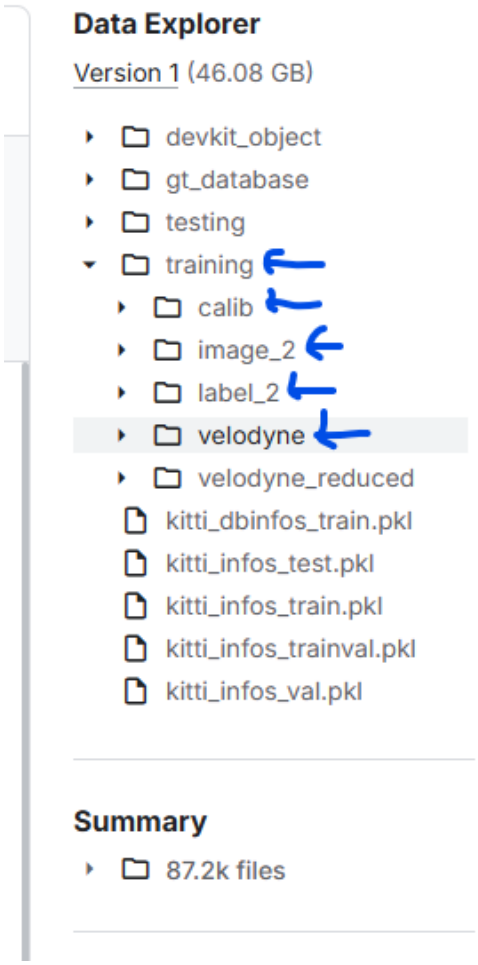


    - Folder Custom-objectDetection-tracking-YOLO contains the following
      - folder train, val, test --> folder containing training, validation and test images with labels
      - installing coder-server.docx --> instructions on how to user UiA HPC for ML tasks
      - instance-segment-tracking-class.ipynb
        - It contains template to detect objects
        - training weights are stored in ./runs/detect/train
        - predictions are stored in ./runs/detect/predict
        - uses vehicle-detection.yaml, yolo11n.pt
      - people-walking.mp4 --> to be used for tracking tasks

After the coder is setup, perform the following tasks

(a) Get familiar with the KITTI data set. There are multiple cameras, and Lidar used. It contains the rich information about - images taken from cameras, point cloud taken from lidar, camera matrices. **Illustrate your understanding in presentation**.

(b) Download data from here. Use the training data set



Perform the lidar-camera fusion. Choose random five images from the training/

- **(i) project them on the lidar frame or viseversa.**
- **(ii) Detect the object in the scene, place bounding box around it, measure the distance of object (shown by color bar) and display near the bounding box**
- **(iii) Validate your estimation of distance with ground truth**

Q4. Autonomous navigation and slip estimation 10 +10 + 5 = 25 points

Webots- https://cyberbotics.com/ is an open source light weight platform to simulate robotics, perception, and control. This give us the platform to conceptualize and simulate downstream tasks.

Leveraging Webots perform the following tasks:

(a) Baseline: Instantiate a robot, and equip it with sensors – camera, lidar, accelerometer, gyro, GPS, IMU etc on it. Then navigate in the world and move towards a predefined object goal (e.g. cylinder sphere or .. with predefined colour) with collision avoidance. At each time record the sensors data e.g. encoder data, lidar data, camera IMU for example. Compute the longitudinal slip (s_{long}) as follows:

$$s_long = \frac{v_{enc} - v_x}{v_x}$$

Where v_{enc} is the velocity obtained from encoder = ωr; ω and r are the angular velocity and radius of wheel.

v_x is the ground truth given from lidar, or the webot simulation world. Feel free to choose

(b) Low slip conditions: Simulate lower slip of your choice (varying the friction), then repeat the same object goal navigation as above. Recalculate the longitudinal slip

(c) Plot the slip for both cases along the trajectory that the robot navigates